Lim & Saloma (2002): equations, variables and definitions

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Field of one focused beam (p. 122, Eq. 1–2)

$$U(u,v) = \frac{2\pi i a^2 A}{\lambda f^2}\, e^{-i(f/a)^2 u} \int_0^1 J_0(v\rho)\, e^{-\frac{i}{2}u\rho^2}\, \rho\, d\rho, \qquad u = \frac{2\pi}{\lambda}\Big(\frac{a}{f}\Big)^2 z, \quad v = \frac{2\pi}{\lambda}\frac{a}{f}\sqrt{x^2+y^2}$$

Paraxial scalar Debye field of a uniformly illuminated, aberration-free lens; $a/f = \mathrm{NA}$ for $n = 1$. In the code: `mcdo.twobeam.beam_field(model='paraxial')`, which also carries the carrier e^{ikz} so that two beams of one color interfere.

Second beam (p. 122–123)

Beam 2 is focused toward $O' = (x_c, z_c)$ by a lens whose axis makes an angle θ with the z axis. Its field at P is the one-beam field evaluated in the beam's own frame:

$$z_b = (x - x_c)\sin\theta + (z - z_c)\cos\theta, \qquad x_b = (x - x_c)\cos\theta - (z - z_c)\sin\theta,$$

a rotation of real space. The paper writes the rotation in optical coordinates, $u_r = u\cos\theta - v\sin\theta$, $v_r = v\cos\theta + u\sin\theta$ (p. 122); that form does not reproduce its Fig. 4 ($-0.21x_c^2$ against $-0.06x_c^2$), and the real-space rotation does.

Signals (p. 123)

$$I_1(P) = |U_1(P)|^2, \qquad I_2(P;\theta;x_c,z_c) = |U_2(P;\theta;x_c,z_c)|^2$$

$$F_{2c}(P;\theta;x_c,z_c) = \kappa_{2c}\, I_2(P;\theta;x_c,z_c)\, I_1(P), \qquad F_{2p}(P;\theta;x_c,z_c) = \kappa_{2p}\, |U_2(P;\theta;x_c,z_c) + U_1(P)|^4$$

The point spread function is each signal normalized to its peak. $d = (x_c^2 + z_c^2)^{1/2}$; confocality holds iff $d = 0$.

Peak fluorescence (p. 126)

F_{2c}^p and F_{2p}^p: peak of the misaligned signal, relative to the aligned peak. Quoted fits:

$$F_{2c}^p(\pi/2) = -0.06x_c^2 + 1, \quad F_{2c}^p(0) = -0.37x_c^4 + 0.01x_c^3 - 1.14x_c^2 + 0.98 \ \text{(valid near } x_c = 0),$$

$$F_{2c}^p(0) = -0.027z_c^2 + 0.996, \qquad F_{2c}^p(\pi/2) = -0.045z_c^2 + 0.986 \qquad (x_c = 0).$$

Linfoot's criteria (p. 126)

Aligned PSF $t = \{t(m)\}$ and misaligned PSF $a = \{a(m)\}$, each normalized to its peak, on a 100×100 x–z grid centered at O; $\langle\cdot\rangle$ is the grid average.

$$F = 1 - \frac{\langle (t-a)^2 \rangle}{\langle t^2 \rangle}, \qquad C = \frac{\langle a^2 \rangle}{\langle t^2 \rangle}, \qquad Q = \frac{\langle |a|\,|t| \rangle}{\langle t^2 \rangle}, \qquad 2Q - C = F = 1 - \epsilon.$$

F fidelity, C structural content, Q correlation quality, ϵ normalized mean-squared error. Perfect recovery: $F = C = Q = 1$. Since $Q \geq 0, F \geq -C$.

Variables

Symbol	Meaning	Value
λ_1, λ_2	Excitation wavelengths, beam 1 and beam 2 (2CE)	750, 656 nm
λ_{2p}	Excitation wavelength, both beams (2PE)	700 nm
λ_e	Single-photon excitation wavelength, $1/\lambda_e = 1/\lambda_1 + 1/\lambda_2$	350 nm
λ_f	Fluorescence wavelength, taken equal to λ_e	350 nm
NA	Numerical aperture, a/f	0.5
n	Refractive index	1
$a,\ f$	Aperture radius and focal length of each lens	—
k	Wavenumber $2\pi/\lambda$	—
$u,\ v$	Axial and radial optical coordinates	—
ρ	Normalized pupil radius	$0 \leq \rho \leq 1$
θ	Angle between the lens axes	$0,\ \pi/2,\ \pi$
$O,\ O'$	Geometric foci of L1 and L2	—
$x_c,\ z_c$	Position of O', in units of λ_1 (or λ_{2p})	$\lvert x_c\rvert \leq 1.35,\ \lvert z_c\rvert \leq 6.6$
d	Focus separation $(x_c^2 + z_c^2)^{1/2}$	—
U_1, U_2	Fields of beams 1 and 2	—
I_1, I_2	Intensities $\lvert U_1\rvert^2, \lvert U_2\rvert^2$	—
F_{2c}, F_{2p}	2CE and 2PE fluorescence	—
κ_{2c}, κ_{2p}	2CE and 2PE cross-sections	—
F_{2c}^p, F_{2p}^p	Peak fluorescence relative to the aligned case	—
$F,\ C,\ Q$	Linfoot fidelity, structural content, correlation quality	—